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Lab section B01

ENCM 335 - LAB 9

Exercise A

```
$ ./a.exe

Success with file name test9A-1.l9l.
List has 6 items ...
  0.25
  -1.5
  -1.75
  2.5
  -0.5
  0

Failure code -6 with file name test9A-2.l9l.

Failure code -5 with file name test9A-3.l9l.

Success with file name test9A-4.l9l.
List has 7 items ...
  11
  0
  0
  12.5
  0
  6.5
  -2.25

Failure code -6 with file name test9A-5.l9l.

Failure code -4 with file name test9A-6.l9l.
```

Files **“test9A-2.l9l”** and **“test9A-5.l9l”** have the failure code **-6** which means that the trailers of those files are invalid.
File **“test9A-3.l9l”** has the failure code **-5** which means that there are not enough data in the file.
File **“test9A-6.l9l”** has the failure code **-4** which means that its header is invalid.
The remaining files are valid and have outputs as shown above

Exercise B

Code:

write_l9l.c

```c
// ENCM 335 Fall 2021 Lab 9 Exercise B
// Implementation of the write_l9l function.

#include <stdio.h>
#include "write_l9l.h"

int write_l9l(const char *filename, const double* a, size_t n)
{
    FILE *fp = fopen(filename, "wb");
    int len=n;
    if (fp == NULL)
        return WL9L_OPEN_FAIL;

    fwrite((void*) &L9L_FIRST_4, sizeof(char), 4, fp);
    fwrite((void*) &len, sizeof(int), 1, fp);
    fwrite((void*) a, sizeof(double), n, fp);
    fwrite((void*) &L9L_FIRST_4, sizeof(char), 4, fp);
    fwrite((void*) &len, sizeof(int), 1, fp);

    if (fclose(fp) != 0)
      return WL9L_CLOSE_FAIL;


    return WL9L_SUCCESS;
}
```

main9B.c

```c
// ENCM 335 Fall 2021 Lab 9 Exercise B
// Program to test use of the write_l9l function.

#include <stdio.h>
#include <stdlib.h>

#include "write_l9l.h"
#include "list_dbl_t.h"
#include "read_l9l.c"

int main(void)
{
  double x[ ] = {1.5, 2.5, 3.5, 4.5, 5.5, 6.5};
  double y[ ] = {-4.5, -3.5, -2.5, -1.5, -0.5, 0.5, 1.5, 2.5, 3.5, 4.5};

  int wl9l_result;
  list_dbl_t ld_object;
  size_t i;

  wl9l_result = write_l9l("test9B-1.l9l", x, sizeof(x) / sizeof(x[0]));
  // INSTRUCTIONS: Replace this comment with code to check the value
  // of wl9l_result and print an appropriate message.

  if (wl9l_result == WL9L_SUCCESS) {
    read_l9l("test9B-1.l9l", &ld_object);
    printf("\nSuccess with file name %s.\n", "test9B-1.l9l");
    printf("List has %zd items ...\n", ld_object.len);
    for (i = 0; i < ld_object.len; i++)
      printf("  %.18g\n", ld_object.arr[i]);
  }
```

```c
      printf("  %.18g\n", ld_object.arr[i]);
  }
  else
    printf("\nFailure code %d with file name %s.\n",wl9l_result,"test9B-1.19l");


  wl9l_result = write_19l("test9B-2.19l", y, sizeof(y) / sizeof(y[0]));
  // INSTRUCTIONS: Replace this comment with code to check the value
  // of wl9l_result and print an appropriate message.

  if (wl9l_result == WL9L_SUCCESS) {
    read_19l("test9B-2.19l", &ld_object);
    printf("\nSuccess with file name %s.\n", "test9B-2.19l");
    printf("List has %zd items ...\n", ld_object.len);
    for (i = 0; i < ld_object.len; i++)
      printf("  %.18g\n", ld_object.arr[i]);
  }
  else
    printf("\nFailure code %d with file name %s.\n",wl9l_result,"test9B-2.19l");

  wl9l_result = write_19l("noSuchDir/noSuchFile.19l",
                         y, sizeof(y) / sizeof(y[0]));
  // INSTRUCTIONS: Replace this comment with code to check the value
  // of wl9l_result and print an appropriate message.

  if (wl9l_result == WL9L_SUCCESS) {
    read_19l("noSuchDir/noSuchFile.19l", &ld_object);
    printf("\nSuccess with file name %s.\n", "noSuchDir/noSuchFile.19l");
    printf("List has %zd items ...\n", ld_object.len);
    for (i = 0; i < ld_object.len; i++)
      printf("  %.18g\n", ld_object.arr[i]);
  }
  else
    printf("\nFailure code %d with file name %s.\n",wl9l_result,"noSuchDir/noSuchFile.19l");

  return 0;
}
```

Output:

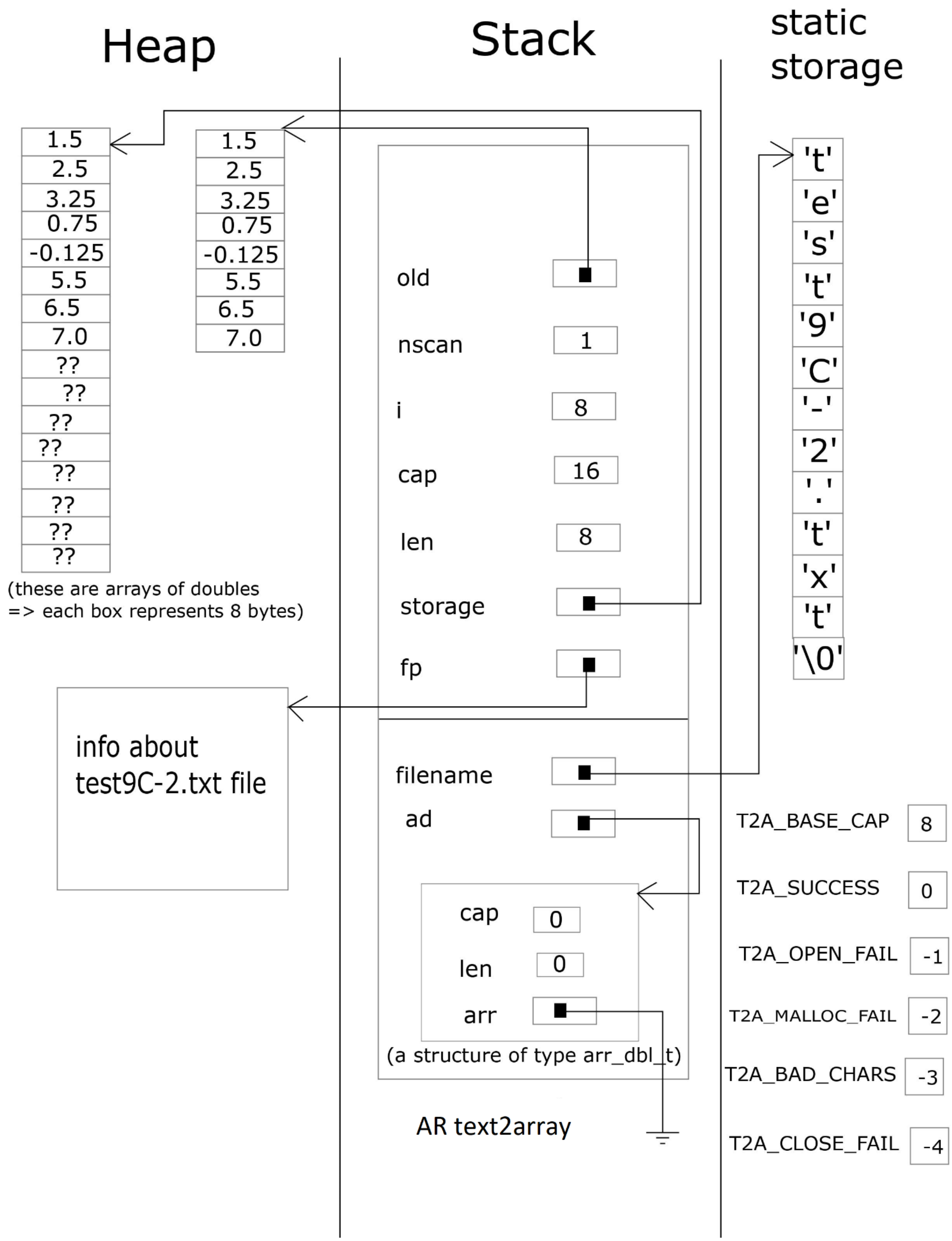
```
$ ./a.exe

Success with file name test9B-1.19l.
List has 6 items ...
  1.5
  2.5
  3.5
  4.5
  5.5
  6.5

Success with file name test9B-2.19l.
List has 10 items ...
  -4.5
  -3.5
  -2.5
  -1.5
  -0.5
  0.5
  1.5
  2.5
  3.5
  4.5

Failure code -1 with file name noSuchDir/noSuchFile.19l.
```

Exercise C


Heap
Stack
static storage
1.5
2.5
3.25
0.75
-0.125
5.5
6.5
7.0
??
??
??
??
??
??
??
??
1.5
2.5
3.25
0.75
-0.125
5.5
6.5
7.0
(these are arrays of doubles => each box represents 8 bytes)
info about test9C-2.txt file
old
nscan 1
i 8
cap 16
len 8
storage
fp
filename
ad
cap 0
len 0
arr
(a structure of type arr_dbl_t)
AR text2array
't' 'e' 's' 't' '9' 'C' '-' '2' '.' 't' 'x' 't' '\0'
T2A_BASE_CAP 8
T2A_SUCCESS 0
T2A_OPEN_FAIL -1
T2A_MALLOC_FAIL -2
T2A_BAD_CHARS -3
T2A_CLOSE_FAIL -4